P1

(a) • 4 sites per unit cell (A/B sublattice for layers 1, 2)

• BZ is same as for single-layer graphene. The underlying Bravais lattice is still triangular w/ same lattice constant.

• Hopping amplitudes of this ~~type~~ range are usually on the eV scale. (The actual value is around 0.3 eV)

(b) We go to momentum space because there is translation symmetry.

Following class notes, we get

$$\boxed{\begin{aligned} H_{top} &= \sum_{\vec{k}\in BZ} \sum_{\sigma=\uparrow,\downarrow} \left[\xi(\vec{k})\, c^{\dagger}_{A\vec{k}\sigma} c_{B\vec{k}\sigma} + h.c. \right] \\ H_{bottom} &= \sum_{\vec{k}\in BZ} \sum_{\sigma=\uparrow,\downarrow} \left[\xi(\vec{k})\, d^{\dagger}_{A\vec{k}\sigma} d_{B\vec{k}\sigma} + h.c. \right] \\ H_{interlayer} &= -t' \sum_{\vec{k}\in BZ} \sum_{\sigma=\uparrow,\downarrow} \left[d^{\dagger}_{A\vec{k}\sigma} c_{B\vec{k}\sigma} + h.c. \right] \end{aligned}}$$

$$\boxed{\begin{aligned} \xi(\vec{k}) = -t \Big[& 2\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{k_y a}{2\sqrt{3}}\right) + \cos\left(\frac{k_y a}{\sqrt{3}}\right) \\ & - i\left(2\cos\left(\frac{k_x a}{2}\right)\sin\left(\frac{k_y a}{2\sqrt{3}}\right) - \sin\left(\frac{k_y a}{\sqrt{3}}\right)\right) \Big] \end{aligned}}$$

(c) $F_\sigma^\dagger(\vec{k}) = \left(c^\dagger_{A\vec{k}\sigma} \quad c^\dagger_{B\vec{k}\sigma} \quad d^\dagger_{A\vec{k}\sigma} \quad d^\dagger_{B\vec{k}\sigma} \right)$ $\qquad F_\sigma(\vec{k}) = \begin{pmatrix} c_{A\vec{k}\sigma} \\ c_{B\vec{k}\sigma} \\ d_{A\vec{k}\sigma} \\ d_{B\vec{k}\sigma} \end{pmatrix}$

$$\Rightarrow \boxed{H = \sum_{\vec{k}\in BZ} \sum_{\sigma=\uparrow,\downarrow} F_\sigma^\dagger(\vec{k})\, \mathcal{H}(\vec{k})\, F_\sigma(\vec{k})}$$

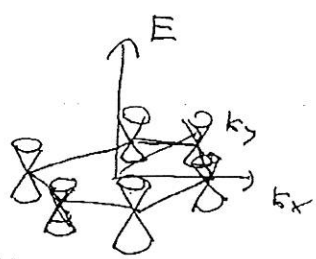
$$\boxed{\mathcal{H}(\vec{k}) = \begin{pmatrix} 0 & \xi(\vec{k}) & 0 & 0 \\ \xi^*(\vec{k}) & 0 & -t' & 0 \\ 0 & -t' & 0 & \xi(\vec{k}) \\ 0 & 0 & \xi^*(\vec{k}) & 0 \end{pmatrix}}$$

(d) Band energies are

$$\boxed{E_{s_1 s_2}(\vec{k}) = \frac{s_1}{2}\left[t' + s_2 \sqrt{(t')^2 + 4|\xi(\vec{k})|^2} \right]}$$

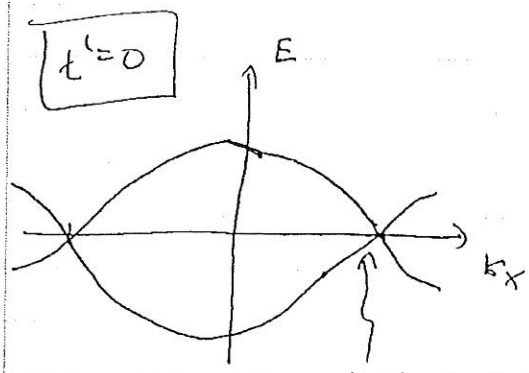
$s_1, s_2 \in \pm 1$

With $t'=0$, plot bands across BZ to find usual Dirac cones:



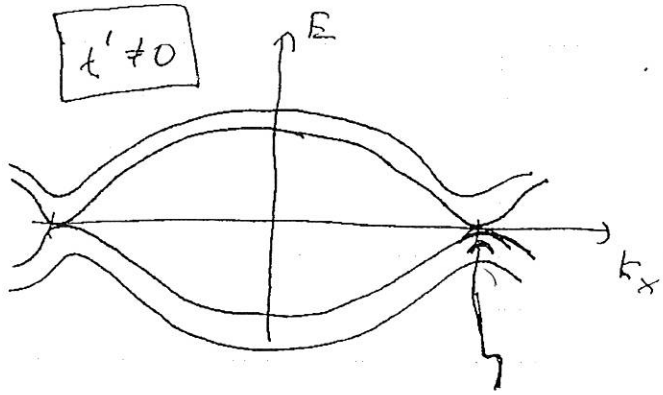
With $k_y = 0$, get

$\boxed{t'=0}$



bands disperse linearly away from Dirac pt.

$\boxed{t' \neq 0}$



bands disperse quadratically at Dirac pts. when $t \neq 0$

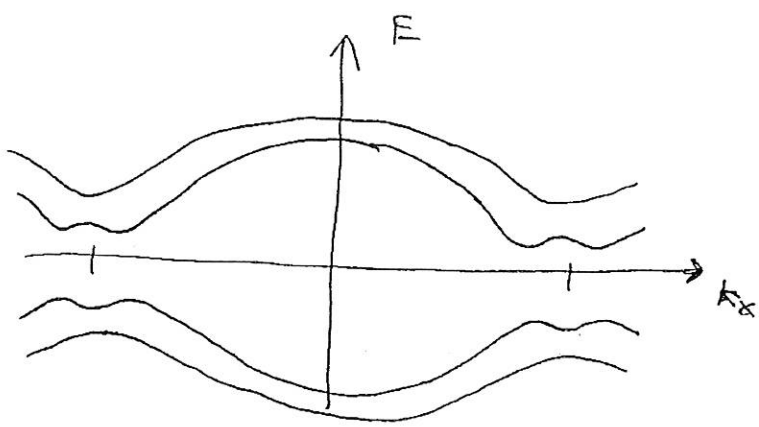
(e) Can write

$$\boxed{\begin{aligned} \delta H &= \sum_{\vec{k}\in BZ} \sum_{\sigma=\uparrow,\downarrow} F_\sigma^\dagger(\vec{k})\, \delta\mathcal{H}(\vec{k})\, F_\sigma(\vec{k}) \\ \delta\mathcal{H}(\vec{k}) &= \begin{pmatrix} \delta_\mu & & & 0 \\ & \delta_\mu & & \\ & & -\delta_\mu & \\ 0 & & & -\delta_\mu \end{pmatrix} \end{aligned}}$$

So the band energies are eigenvalues of

$$\mathcal{H}(\vec{k}) + \delta\mathcal{H}(\vec{k}).$$

A band gap develops when $\delta_\mu \neq 0$, and the spectrum then looks like this:



$\Rightarrow$ System becomes an insulator!